Class Activity # 6
Calculus Physics 2

ıme: Key Date: _____

Group Members: _____ _____ _____

Solve the following problems together as a group, stay engaged in the conversation with the group!

➢ Use **Chapter 53 and 57 Lecture Notes** as a guide to answer the following questions.

1) Which of the following statements about electromagnetic waves in vacuum are true? (There could be more than one correct choice.) (B) (C) (D) (F)

A) The higher-frequency EM waves travel faster than the lower-frequency EM waves in vacuum.
(B) The higher-frequency EM waves travel at the same speed as the lower-frequency EM waves in vacuum.
(C) The higher-frequency EM waves have shorter wavelengths than the lower-frequency EM waves.
(D) The electric field vector is always at right angles to the magnetic field vector.
E) The electric field vector does not have to be at right angles to the magnetic field vector.
(F) The lower the frequency of the EM wave the lower the energy.
G) The energy of an EM waves is the same for all frequencies.

2) Which type of wave can travel through vacuum?

A) Mechanical Waves
(B) Electromagnetic waves
C) Water waves
D) All the above

3) The time for one cycle of a wave is known as the

(A) Period of the wave.
B) Wavelength of the wave.
C) Amplitude of the wave.
D) Velocity of the wave.
E) Frequency of the wave.

4) The distance a wave travels in one cycle is known as the

A) Period of the wave.
(B) Wavelength of the wave.
C) Amplitude of the wave.
D) Velocity of the wave.
E) Frequency of the wave.

5) The speed (in meters per second) of all EM waves in vacuum is equal to $c = 3 \times 10^8$ m/s

6) A light-year is a measure of

A) Time.
(B) Distance.
C) Wavelength.
D) Period.

7) The Hubble telescope detects a galaxy that is 15,000 light years away, how much time did it take the light from that galaxy to reach the telescope in years?

15,000 years

8) Which portion of the visible do the photons have the highest energy?

(A) Blue
B) Yellow
C) Red
D) Green
E) They all have the same energy.

9) If a star is red shifted, what is the star doing relative to Earth?

A) Coming towards Earth.
B) Not moving relative to Earth.
(C) Moving away from Earth.
D) Can not be determined.

moving away
$+2 \times 10^7$ m/s

10) Mike the astronomer knows that a galaxy is moving away from the earth at $v = 2 \times 10^7$ m/s, if the frequency of hydrogen is line used by Mike is 1.42 GHz measured on earth, determine the frequency seen by Mike for the galaxy.

$$f = f' \sqrt{\frac{1+\frac{v}{c}}{1-\frac{v}{c}}} = f' \sqrt{\frac{0.933}{1.0667}} = f'(0.975) = 1.336\,\text{Hz}$$

11) Rank the EM waves in order from lowest energy (1) to highest energy (7)

Ultraviolet waves	5
X-rays	6
Visible light	4
Microwaves	2
Gamma rays	7
Radio waves	1
Infrared radiation	3

12) What EM wave is used by your cell phone to communicate with the cell towers?

A) infrared light.
B) gamma rays.
C) visible light.
D) ultraviolet light.
(E) microwaves
F) radio waves

13) Bob walks into a dark room in a haunted house, he notices that the white in his shirt is glowing, what type of EM waves is Bob experiencing?

A) infrared light.
B) gamma rays.
C) visible light.
(D) ultraviolet light.
E) microwaves
F) radio waves

14) Cathy puts her hand over the burner on a stove and feels the warmth, Cathy is "feeling" what form of energy?

A) Ultra volet radiation.
(B) Infrared radiation.
C) Visible light.
D) Gamma radiation.
E) None of the above.

15) Identify the source of the portions of the EM spectrum below.

a. Molecular vibrations
b. The nucleus of the atom.
c. Electron transitions in the atom. Energy
d. Very high electron transitions in the atom.
e. accelerating electrons.

Radio waves	e
Microwaves	e
Infrared radiation	a
Visible light	c
Ultraviolet light	c → d
X-rays	d
Gamma rays	b

16) Convert the following (assume EM waves are in vacuum) $c = 3 \times 10^8$ m/s

A) Blue light has a wavelength of 420 nm, what is its frequency? $nm = 10^{-9} m$

$c = f\lambda \quad | \quad f = \frac{c}{\lambda} = 7.14 \times 10^{14}\ Hz$

B) A Wifi operate at 5 GHz, what is its wavelength? $GHz = 10^9\ Hz$

$c = f\lambda \quad | \quad \lambda = \frac{c}{f} = 0.06m$

17) The value of the rms electric field for a certain type of electromagnetic wave is 570 N/C. What is the value of the rms magnetic field for that wave?

Peak $\vec{E}_{rms} = \frac{\vec{E}_p}{\sqrt{2}} = 403\ N/C$ | $|\vec{B}| = \frac{|\vec{E}|}{c} = 1.34 \times 10^{-6}\ T$

18) Given an EM wave with an intensity of $I = 200\ W/m^2$,

A) What is the magnitude of the rms electric field?

$I = c\varepsilon_0 E^2$ | $E = \sqrt{\frac{I}{c\varepsilon_0}} = 274.5\ N/C$

B) What is the magnitude of the rms magnetic field?

$I = \frac{cB^2}{\mu_0}$ | $B = \sqrt{\frac{\mu_0 I}{c}} = 9.15 \times 10^{-7}\ T$ | or $B = \frac{E}{c} = 9.15 \times 10^{-7}\ T$

19) A 20 W light source radiates uniformly in all directions and can be considered a point source.

A) What is the magnitude of the electric field at 2 m from the light?

$I = \frac{P}{4\pi r^2} = c\varepsilon_0 E^2$ | $E = \sqrt{\frac{P}{4\pi r^2 \varepsilon_0 c}} = 12.24\ N/C$

B) A detector measures the magnitude of the electric field at some distance from the light is 10 N/C, how far is the detector from the light source?

$\frac{P}{4\pi r^2} = c\varepsilon_0 E^2$ | $r = \sqrt{\frac{P}{4\pi c \varepsilon_0 E^2}} = 2.45\ m$

20) If a solar cell with an area of $2\ m^2$ receives $800\ W/m^2$ of light Intensity, the solar cells are 15 % efficient at converting the suns power to electrical power,

A) How much power is converted to electrical power if the cells are at an angle of 0° measured normal to the surface of the cell?

$I = \frac{P}{A}$ | $P = (IA\cos(\theta))\eta = (1600\ W)(0.15) = 240\ W$

B) How much power is converted to electrical power if the cells are at an angle of 50° measured normal to the surface of the cell?

$P = (IA\cos(\theta))\eta = (1028\ W)(0.15) = 154.2\ W$

21) The heating element in Bob's toaster oven emits electromagnetic waves with a peak wavelength of 630 nm, what is the temperature of the heating element? ($1\ nm = 10^{-9} m$)

$\lambda_P = \frac{2.898 \times 10^{-3}\ K\,m}{T}$ | $T = \frac{2.898 \times 10^{-3}\ K\,m}{630 \times 10^{-9}\ m} = 4600\ K$

22) A person blowing glass heats a spherical piece of glass so that it glows red ($T_g = 2{,}000\ °C = 2{,}273\ K$) in a room that has a temperature of $T_{room} = 27\ °C = 300\ K$. The surface area of the glass sphere is $A = 0.3\ m^2$.

A) What is the peak wavelength emitted by the glass sphere when it is glowing red? near IR

$\lambda_P = \frac{2.898 \times 10^{-3}\ K\,m}{T} = 1.27 \times 10^{-6}\ m = 1.27\ \mu m$

B) What is the amount of power emitted in the form of EM radiation by the glass sphere? (the emissivity of glass is $e = 0.9$, and $\sigma = 5.67 \times 10^{-8}\ J/(s m^2 K^4)$)

$P_{net} = \sigma e A (T^4 - T_0^4) = 408{,}520\ W$

23) If unpolarized light of intensity I_o passes through an ideal polarizer, what is the intensity of the emerging light?

A) 0.81 I_o
B) 0.19 I_o
C) 0.22 I_o
D) 0
(E) 0.50 I_o

$I = \frac{1}{2} I_0$ For unpolarized light through a polarizer

24) Unpolarized light of intensity I_o passes through a polarizer that has its polarization axis vertical, what, in terms of the initial intensity I_o, what is the intensity of the light that passes through a second polarizer that has its polarization axis horizontal?

A) 0.81 I_o
B) 0.19 I_o
C) 0.22 I_o
(D) 0
E) 0.50 I_o

Crossed polarizers

25) Unpolarized light with an intensity of I = 1,000 W/m^2 is incident on a polarizer with its transmission axis oriented 20° from the vertical,

A) What is the intensity of light through the first polarizer (I_1)?

$I_1 = \frac{1}{2} I_0 = 500\ W/m^2$ at 20° to vertical

B) The light then goes through a second polarizer with its transmission axis oriented 60° to the vertical in the same direction as the first, what the intensity of light through the second polarizer (I_2)?

$\theta_2 = 60° - 20° = 40°$ | $I_2 = I_1 \cos^2(40°) = 294.4\ W/m^2$

C) The light encounters a third polarizer with its transmission axis at 40° to the vertical, what is the intensity of light through the third polarizer (I_3)?

$\theta_3 = 60° - 40° = 20°$ | $I_3 = I_2 \cos^2(\theta_3) = 259\ W/m^2$

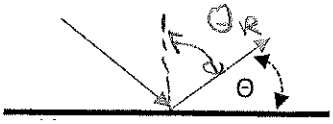
26) The model that uses the fact that light travels in a straight line is known as

A) Tracing Model.
B) Line Model.
(C) The Ray Model.
D) The Wave Model,

27) The reflection off a rough surface is known as

A) Specular Reflection.
(B) Diffuse reflection.
C) All the above.
D) None of the above.

28) A ray of light reflects off the surface shown, it the angle measured from the surface of the reflected beam is Θ = 30°, what is the angle of reflection?



$\theta_R = 90° - 30° = 60°$

always measured from the normal line

29) Light travels from one place to another along a path of least

A) distance.
B) time.
C) effort.
D) complication.

30) When light is absorbed by a material, what is the energy in the light transformed into?

A) Diffuse waves
B) Reflected waves
C) Diffracted waves
D) EM waves
E) Thermal energy

31) If white light is incident on a piece of paper and the colors that are absorbed are Red, Orange, Green, Blue, Indigo and Violet, what color does the paper appear?

A) All the color that are absorbed mixed.
B) Black
C) Yellow
D) White

32) Which type of scattering of light creates the blue sky?

A) Rayleigh Scattering
B) Mie Scattering
C) They both do not depend on the frequency of light.

33) A fire hydrant casts a shadow due to streetlight, if the shadow that is formed is only an umbra, what do you know about how the streetlight acts?

A) It acts as a point source.
B) It acts as a extended source.
C) It could act as either a point source or an extended source depending on how close it is to the fire hydrant.
D) None of the above.